FORAD 2.0 · FROM THE PLAYER MANUAL, CHAPTERS 2–3

How students move through FORAD

One period from start to finish: what happens outside the program, the order a team should work in, every menu, screen and input the manual describes, and what each results report shows. Hover or tap a step to highlight where it happens on the menu map.

🟦 Team works here 🟫 Instructor (FORAD Administrator) ☐ Screen or input in the program

The period loop

Every period runs through four stages (manual §2.1). Only the Decision stage happens inside the program on the team's computer.

INSTRUCTOR
Start
All teams inherit the same Period 0 company. Each period the instructor publishes market commentary and the new assumptions.
`Assumptions file · market report`

→

TEAM
Decision
Predict, decide, update and stress test inside the program until the projected results are acceptable.
`Steps 1–10 below`

→

INSTRUCTOR
Simulation
Your predictions are replaced by actual market data, every team is recalculated and the tender auction is cleared.
`Cartel Summary for all teams`

→

INSTRUCTOR
Outcome
Teams get their updated decisions file and the next assumptions. Last period is closed; the new one opens.
`DecisionsT#P#.txt returned`

next period starts at Decision again

The student path through one period

The manual's five decision steps (Predictions, Operations, Investing, Financing, Hedging) with the checks around them. Menu paths are shown as they appear in FORAD 2.0.

1 Read the market and last period's results
Read the instructor's commentary, check this period's base interest rates, and see how the closed period turned out: every report shows the closed period next to the current projection.
`Decisions › Market Info` `Results › Summary` `Results › PE Scoring` `Results › Consolidated BS · IS` `Cartel Summary`

2 Enter predictions
End-of-period ¥/$ and $/€ on the US Holding hedging tabs; oil price and Japan's sales units and price on Japan Operations; Germany's sales units and price on Germany Operations. Defaults are last period's actuals.
`United States › Hedging JPY / EUR` `Japan › Operations` `Germany › Operations`

3 Germany operations first
Capacity, production, terms, export and inter-company currency, transfer price and dividend to the holding company. German production sets Japan's inter-company sales, so the manual recommends starting here. Open the Tender Market window to bid when a tender is offered.
`Germany › Operations` `Tender Market button` `Ctrl+G`

4 Japan operations
Capacity and production (watch ending inventory), collection and payment days, export currency, dividend to the holding company, oil futures.
`Japan › Operations` `Ctrl+J`

5 Investing
Marketable securities in each unit, inter-company loans from the subsidiaries to the US, share repurchase and dividend per share at the holding company.
`Japan / Germany / United States › Investing`

6 Financing
Check projected cash on days 0, 30, 45, 60 and 90 at the bottom of each screen, then borrow short or long term, lend inter-company, set swaps, call the convertible or issue shares.
`Japan / Germany / United States › Financing` `Swaps buttons`

7 Hedging
Review exposures by currency, then add forwards and options; check the expected FX result against the 100%-forward benchmark and the FX graphs. Interest-rate futures sit on the US hedging tabs.
`each unit › Hedging (two currencies)` `FX graphs button`

8 Update and check the projected reports
Refresh the model, then work down the Results menu: Summary and PE Scoring for the share price; each subsidiary's balance sheet, income statement and cash flow for cash, debt and inventory; Hedging for the FX result; Taxes for the effective rate. See the report guide below for what each one holds.
`Update · Ctrl+R` `Results › Summary · PE Scoring` `Results › Subsidiary › BS · IS · CF` `Results › Taxes`

9 Stress test
Change your predictions to extreme values, update, and ask: is the result acceptable, does the hedge work as planned, what should change? Then go back to steps 3–7.
`Prediction cells` `Update` `Results`

REPEAT 3 – 9 UNTIL READY

10 Save and submit
Save the decision, optionally export the results to CSV, and send the decisions file to the instructor before the deadline.
`File › Save Decision` `File › Save Output Data` `DecisionsT#P#.txt`

→ Instructor runs the period
Actual rates, prices and volumes replace the forecasts. Results come back with the next period's assumptions and commentary.

Shortcuts from the manual: Ctrl+J Japan, Ctrl+G Germany, Ctrl+U United States, Ctrl+R update; type 2.5m for 2,500,000 and 4b for 4,000,000,000.

Menu map: every function

FORAD 2.0's main menu with its screens and inputs (manual §2.2 and chapter 3). White cells are inputs; everything else is calculated and refreshes on Update.

File | Decisions | Update | Results | Window | Help

FORAD 2.0 main menu

File

Save Output Data
Writes Decision and Results Output.csv for Excel

Save Decision
Saves all inputs and refreshes results

Exit

Update `Ctrl+R`

Refresh every calculation
Run after any change; result screens can stay open

Results

Consolidated Balance Sheet `Ctrl+B`

Consolidated Income Statement `Ctrl+I`

PE Scoring
Six P/E elements, EPS, share price

Summary
Opening screen: all three units and consolidated

US Holding Company ›
Statements in USD, Investing, Financing, Hedging JPY / EUR

Japanese Subsidiary ›
- Balance Sheet · JPY / USD
- Income Statement · JPY / USD
- Cash Flow · JPY
- Operations · Investing · Financing
- Hedging · USD / EUR

German Subsidiary ›
Same reports in EUR / USD; Hedging USD / JPY

Taxes ›
Tax summary · dividend and interest gross-up

Window · Help

Switch between open screens
Help is not described in the manual

Decisions

Market Info Base interest rates for the prior and current period, before company credit spreads

Tab	Japan `Ctrl+J`	Germany `Ctrl+G`	United States `Ctrl+U`
Operations Japan and Germany only	• Capacity and production • Sales forecast: domestic, export units, price • Export currency of invoice • Collection and payment days • Dividend to holding company • Oil price forecast • Oil futures (1,000 bbl)	• Capacity and production • Sales forecast: domestic, export units, price • Export currency of invoice • Inter-company price, currency, days • Collection and payment days • Dividend to holding company `Tender Market window: bid price and units, expected result`	*No operations. Royalty income is automatic.*
Investing	• Marketable securities, invest or redeem, beginning or mid period • Inter-company loan to the US and its rate	• Marketable securities, invest or redeem, beginning or mid period • Inter-company loan to the US and its rate	• Marketable securities • Share repurchase (from Period 2) • Dividend per share
Financing Cash on days 0, 30, 45, 60, 90 at the bottom	• Short-term loans in ¥, $, €, beginning or mid • Inter-company borrowing from the US • Long-term fixed and floating debt `Swaps buttons`	• Short-term loans in €, $, ¥, beginning or mid • Inter-company borrowing from the US • Long-term fixed and floating debt `Swaps buttons`	• Short-term loans in $, ¥, € • Inter-company loans to Japan and Germany • Long-term fixed and floating debt • Call the zero-coupon convertible • Issue new shares `Swaps buttons`
Hedging, currency 1	USD $ • Exposures by type • Forwards, call and put options • Expected FX result vs benchmark `FX graphs button`	USD $ • Exposures by type, incl. tender sales • Forwards, call and put options • Expected FX result vs benchmark `FX graphs button`	JPY ¥ • **End-of-period ¥/$ forecast** • Dividend and interest exposure • Forwards, call and put options • Interest-rate futures
Hedging, currency 2	EUR € • Same tools as currency 1	JPY ¥ • Same tools as currency 1	EUR € • **End-of-period $/€ forecast** • Same tools as currency 1 • Interest-rate futures

Predictions live in two places. FX rates on the US hedging tabs; oil and sales on the Operations tabs. Japan and Germany show the FX forecast read-only.

Germany drives Japan. Each German unit uses two Japanese units, so German production sets Japan's inter-company sales.

Every screen shows last period. Most screens put the prior period's actual next to the current decision.

Results reports: what each one shows

Line items as laid out in the FORAD 2.0 program files. Every report shows the closed period next to the current projection, and refreshes on Update.

Balance sheet · subsidiary

`Results › Japanese / German Subsidiary › Balance Sheet`

`local · USD`

CURRENT ASSETS
Cash · Marketable securities · Inter-company loan · Accounts receivable · Inter-company receivable · Inventory

LONG-TERM ASSETS
Gross fixed assets · Accumulated depreciation · Net fixed assets · Other assets · Deferred charges

CURRENT LIABILITIES
Accounts payable · Inter-company payable · Other payables · Overdraft · Inter-company debt · Short-term debt

LONG-TERM LIABILITIES
Floating-rate debt · Fixed-rate debt

EQUITY
Capital stock · Other comprehensive income · Cumulative translation adjustment · Retained earnings

The USD version is the translated balance sheet and adds accrued taxes, APIC and treasury stock lines.

Income statement · subsidiary

`Results › Subsidiary › Income Statement`

`local · USD`

REVENUE
Domestic · Export · Inter-company · Total revenues

OPERATING
Cost of goods sold · SG&A · Depreciation · Operating income

NON-OPERATING
FX gain/(loss) · Oil gain/(loss) (Japan) · Other, net · EBIT

INTEREST AND TAX
Interest income · Interest expense · EBT · Income taxes · Withholding taxes · Net income

BELOW THE LINE
Dividends paid to the holding company

Cash flow · subsidiary

`Results › Subsidiary › Cash Flow`

`local currency only`

OPERATING
Net income · Depreciation · Deferred charges · Change in receivables, inter-company receivable, inventory, payables, inter-company payable, other payables

INVESTING
Fixed assets (capacity) · Marketable securities · Inter-company loan to parent

FINANCING
Overdraft · Short-term debt · Inter-company debt · Fixed-rate debt · Floating-rate debt · Dividends paid

CASH
Net change · Beginning cash · Ending cash

US Holding Company statements

`Results › US Holding Company` `USD`

BALANCE SHEET
Cash · Marketable securities · Inter-company loans to subs · Investments in foreign subs · Overdraft · Short- and long-term debt · Zero-coupon convertible · Capital stock, APIC, treasury stock · Retained earnings

INCOME STATEMENT
Third-party royalties · Foreign-source income: inter-company dividends and interest · SG&A · FX gain/(loss) · Interest · Taxes · Net income before subs · Equity in earnings of subs · Net income

CASH FLOW
Zero-coupon accretion · Loans to subsidiaries · Convertible issue/(repayment) · Share issue and repurchase · Dividends paid

Consolidated statements

`Results › Consolidated Balance Sheet · Ctrl+B`

`Consolidated Income Statement · Ctrl+I`

BALANCE SHEET
Same lines as a subsidiary, without any inter-company items · adds zero-coupon convertible, APIC, treasury stock and CTA

INCOME STATEMENT
Domestic and export revenues only (inter-company sales eliminated) · COGS through net income as for a subsidiary

A consolidated cash flow statement (with the effect of FX changes on cash) exists in the program files but is not on this menu.

Operations report

`Results › Subsidiary › Operations`

VOLUME (UNITS)
Capacity · Beginning inventory · Production · Sales · Ending inventory · Capacity increase · Sales from current production

COST PER UNIT
Inter-company (Germany) · Domestic materials · Imported materials · Oil (Japan) · Labor and overhead · Excess production cost · Basic and total cost per unit

INVENTORY VALUE
Beginning · Production · Sales · Ending

Investing report

`Results › Subsidiary / US › Investing`

INTEREST EARNED
Marketable securities, beginning and mid period · Inter-company loans, beginning and mid period

FIXED ASSETS (SUBSIDIARIES)
Beginning · Capex · Ending · Accumulated depreciation · Net

US ONLY
Loans to Japan and Germany · Interest-rate hedging result

Financing report

`Results › Subsidiary / US › Financing`

DEBT
Short-term, beginning and mid · Weighted average overdraft · Inter-company debt · Fixed- and floating-rate long-term

INTEREST
Short-term · Overdraft · Inter-company · Long-term fixed and floating · Net swap payments · Amortized issue costs · Total interest

RATES AND SPREADS
Current short- and long-term credit spread · Historical long-term spread and fixed rate

Hedging report · per currency

`Results › Subsidiary / US › Hedging`

EXCHANGE RATES
Beginning · Forward · End of period (your forecast) · Volatility

EXPOSURE BY TYPE
Export sales · Inter-company · Imported materials · Tender revenue · Dividend · Loans · Investments · Total

HEDGES
Forwards · Calls and puts: notional, strike, premium

OUTCOME
Result on exposures and on hedges · Total net FX gain/(loss) · Forward benchmark vs actual · % variance to benchmark

PE Scoring

`Results › PE Scoring`

CREDITWORTHINESS 30%
Debt to assets · Times interest earned · Short-term debt to working capital · Previous and new credit rating

CURRENCY HEDGING 20%
% FX variance to forward

PLANNING 10%
EPS actual vs forecast

TAX MANAGEMENT 20%
Effective tax rate

DIVIDEND MANAGEMENT 10%
EPS and dividend growth · Cut penalty · Payout · Volatility

EARNINGS QUALITY 10%
Non-core earnings share · EPS volatility · Dilution

OVERALL
Change in P/E · P/E · EPS · Annualized EPS · Stock price

Taxes

`Results › Taxes`

TAX SUMMARY
EBT, income tax, withholding, tax on foreign-source income, total and effective rate for US, Japan, Germany and consolidated · Foreign tax credit balance: beginning, Japan, Germany, ending

DIVIDEND GROSS-UP (JAPAN, GERMANY)
Dividend before and after withholding · USD value · Deemed-paid foreign taxes · Grossed-up dividend · Theoretical US tax · Foreign tax credit · Additional US tax · Excess credit/(deficit) · Net dividend after tax

INTEREST GROSS-UP (JAPAN, GERMANY)
Same steps for inter-company interest

Also in the program

`Summary · Tender · IC eliminations`

SUMMARY
Opening screen: projected results for all three units and consolidated

TENDER SALES WINDOW
Market size · Bid price and units · Expected price and units · Tender revenue

INTER-COMPANY ELIMINATIONS
Japan's inter-company profit per unit · Profit left in Germany's COGS and inventory · Eliminating entries

How the reports tie together

- Each balance sheet balances: total assets equal liabilities plus equity.
- Ending inventory units on the Operations report match the Operations decision screen.
- The Hedging report's total gain/(loss) is the FX line on the income statement; oil futures land in oil gain/(loss).
- The effective rate in the Tax summary is the input to the Tax management score in PE Scoring.
- Ending cash on a cash flow statement is the cash line on that unit's balance sheet.
- Inter-company revenue, receivables, payables and loans disappear on consolidation.
- Subsidiary dividends leave in their cash flow and arrive as foreign-source income in the US income statement.

Where each part lives in the FORAD web app

How the redesign regroups the same functions.

FORAD 2.0	Web app
`Commentary, Market Info`	Top bar **Period N market report** pop-up, and Menu › Market Reports & Results
`Decisions › Japan / Germany / US`	Decision workspace: country tabs (US Holding, Japan, Germany), each with Operations, Investing, Financing, Hedging
`Tender Market window`	Germany › Operations, below the operations table (S-07)
`Update · Ctrl+R`	Not needed: projections recalculate as you type
`Results › Summary, PE Scoring`	Live projection and Review & submit; closed periods under Menu › Results
`Results › statements by unit`	Results tab, by entity: balance sheet, income statement, cash flow, operations, investing, financing, hedging, taxes (S-16, not built yet)
`Stress testing (manual §2.2)`	Stress test lab, from the dashboard
`File › Save Decision, email the .txt`	Autosaved draft; Submit in Review & submit
`Cartel Summary`	Results › stock price chart and leaderboard

Source: FORAD 2.0 Player Manual (Tower46), §2.1 Simulation Overview, §2.2 Model Navigation, §2.2 Stress Testing, chapter 3 Decisions Making, FORAD Tips 2.1 and 3.4, and the report layouts in the v2.1 program files (statement definition files and Results menu).